

THE STUDENT PORTAL

a unified academic information system

CAPITAL UNIVERSITY · GRADUATION PROJECT · 2026

FACULTY EXAMINATION COPY · §1.0



FIG. I — THE SEAL

CONTENTS

I



INTRODUCTION
PROBLEM & OBJECTIVES

II



LITERATURE
RELATED WORK

III



ARCHITECTURE
MODULAR MONOLITH

IV



DOMAIN
CQRS · HIERARCHY · DAG

V



AUTHORITY
SCOPE-FIRST ACCESS

VI



SYNCHRONY
SIS INTEGRATION

VII



INTERFACE
TWO PORTALS · RTL

VIII



EVALUATION
SCALE & CONCLUSION

ONE BUILDING, MANY ROOMS

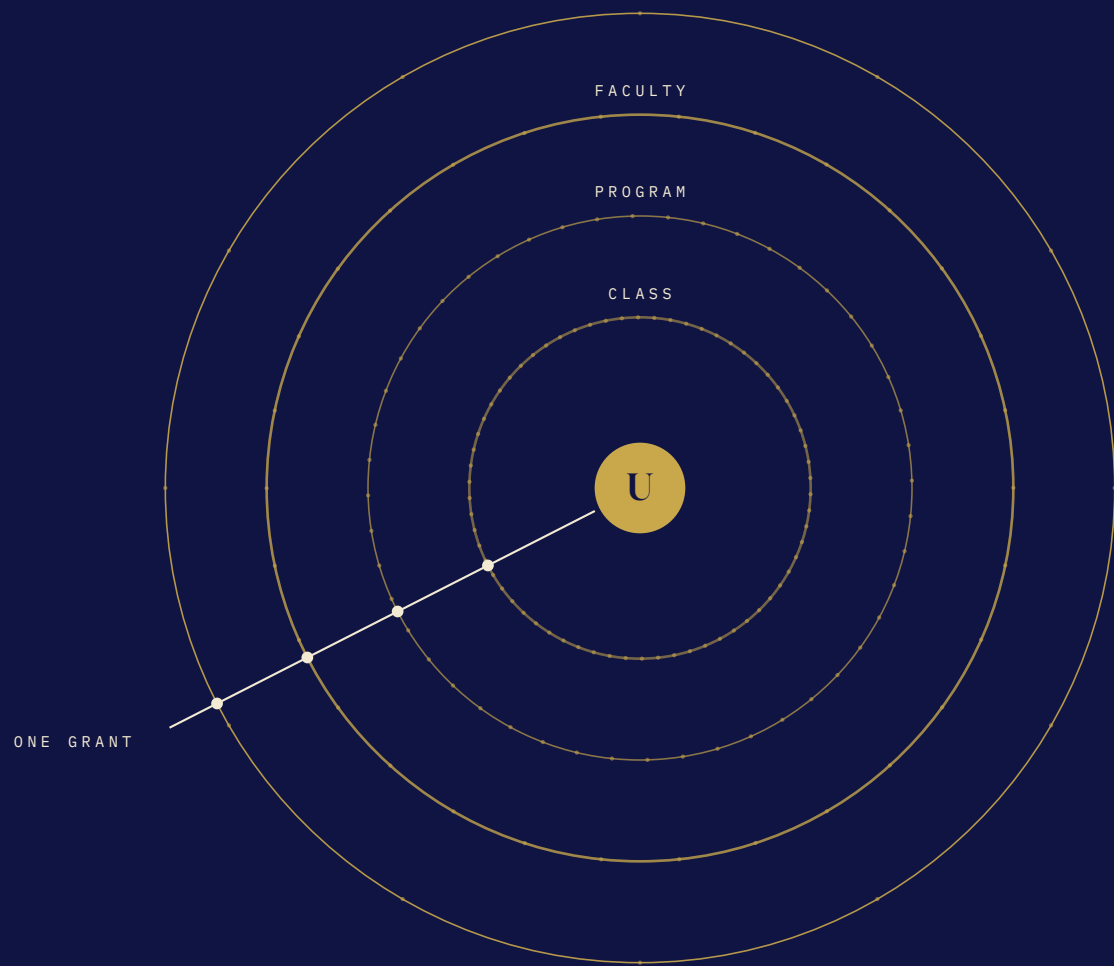
FIG. II — THE MODULAR MONOLITH

§3.1



AUTHORITY IS A PLACE

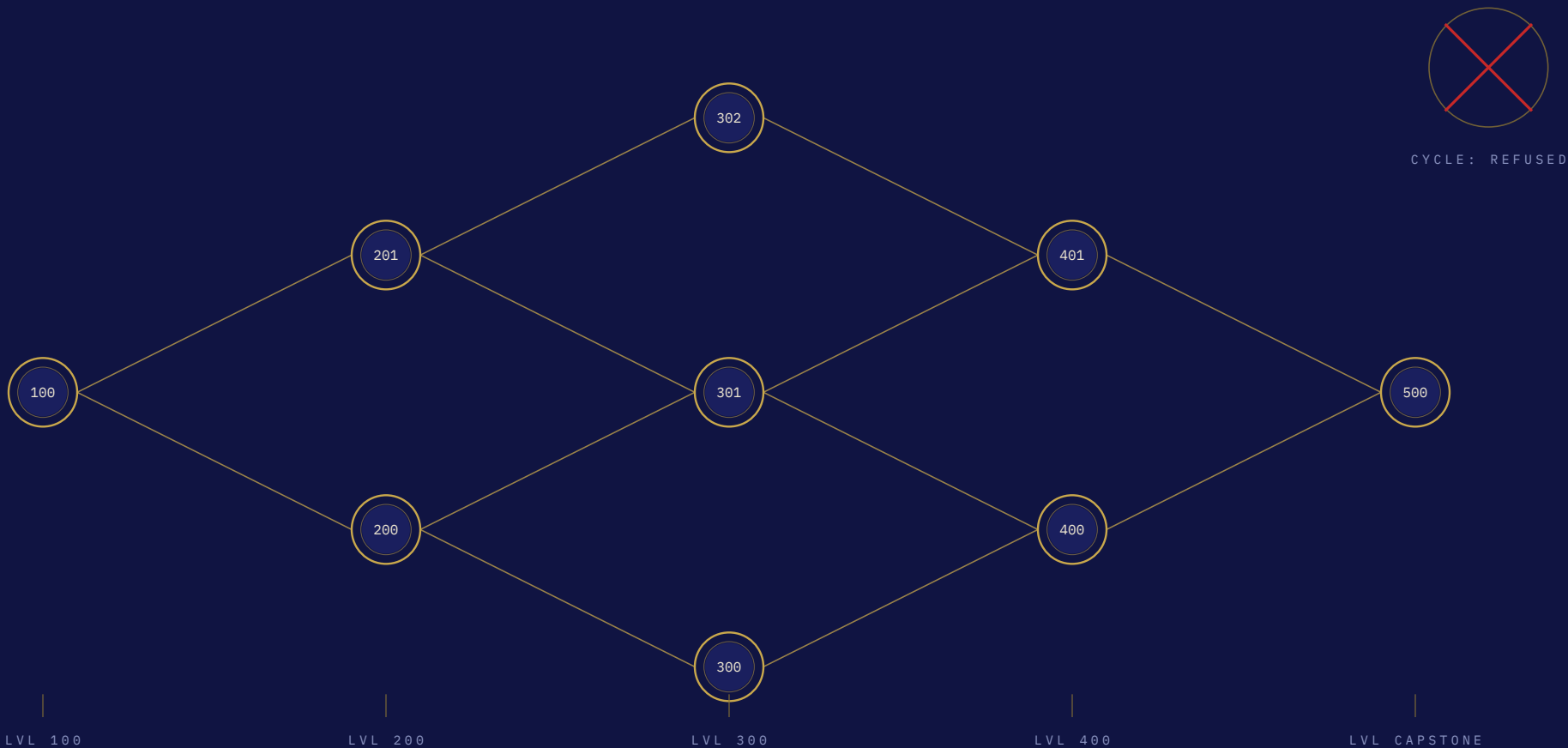
FIG. III — SCOPE-FIRST AUTHORIZATION · PERMISSION × SCOPE
DOMAIN



MODULE.ENTITY.ACTION @ NODE

NOTHING BEFORE ITS TIME

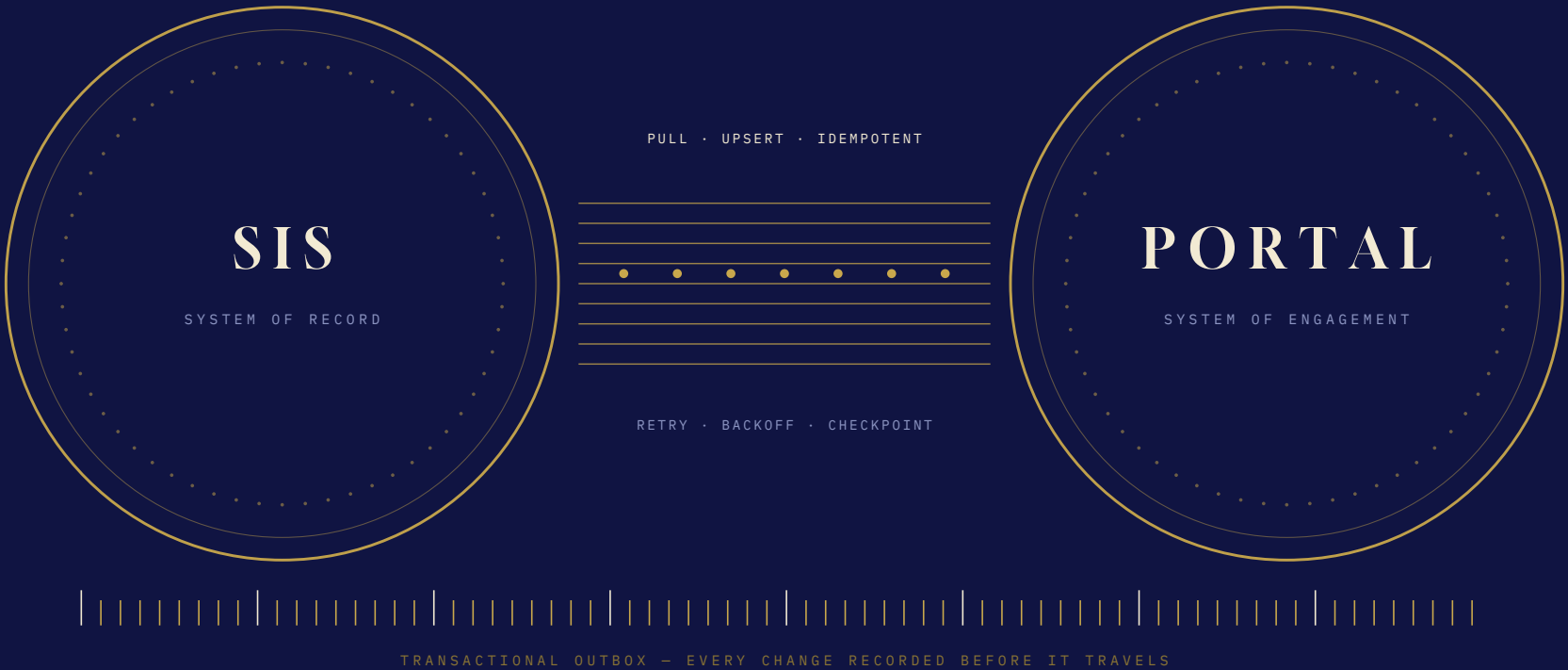
FIG. IV - PREREQUISITE GRAPH - ACYCLIC BY PROOF



DEPTH-FIRST SEARCH · $O(V+E)$

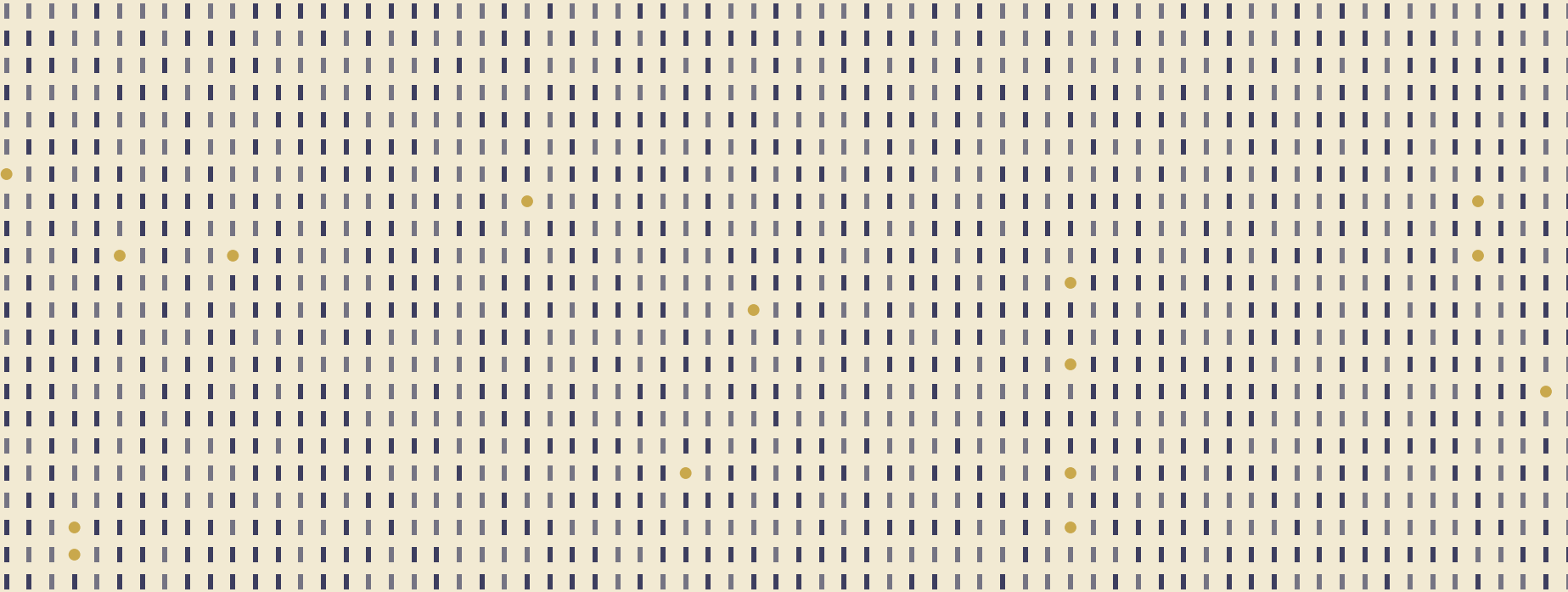
TWO LEDGERS, ONE TRUTH

FIG. V — SYNCHRONIZATION WITH THE SYSTEM OF RECORD



THE CENSUS

FIG. VI — ONE MARK, ONE RECORD · SEEDED AT FULL SCALE



12,400

STUDENTS

118,000

REGISTRATIONS

96,500

GRADE ENTRIES

<180 ms

API P95

PLATE VII · EVALUATION & SCALE



what a university knows, made navigable

CAPITAL UNIVERSITY · STUDENT PORTAL · MMXXVI

SUBMITTED IN PARTIAL FULFILMENT · EXAMINED WITH GRATITUDE